

Quiz (Carolina Reaper)

- Q1.** A shipping company's delivery times t (in hours) versus distance d (in miles) are related by $t = \frac{1}{40}d + 2$. What is the slope?
(A) 0.025
(B) 0.04
(C) 0.1
(D) 2
(E) 40
- Q2.** The graph of $y = mx + b$ has a slope of -3 and a y -intercept of $(0, 5)$. What is the equation of the line?
(A) $y = -3x + 5$
(B) $y = 3x - 5$
(C) $y = -3x - 5$
(D) $y = 3x + 5$
(E) $y = -5x - 3$
- Q3.** A traffic flow model predicts the number of cars n passing a checkpoint in t minutes as $n = 2t + 5$. What is the rate of change of cars per minute?
(A) 1 car/min
(B) 2 cars/min
(C) 3 cars/min
(D) 4 cars/min
(E) 5 cars/min
- Q4.** What is the slope of the line that passes through $(2, 3)$ and $(4, 5)$?
(A) 0.5
(B) 1
(C) 1.5
- (D) 2
(E) 0
- Q5.** The table shows the distance traveled by a truck over time. What is the slope?
(A) -10
(B) -5
(C) 0
(D) 5
(E) 10
- Q6.** A logistics company's cost C (in dollars) to transport x packages is given by $C = 2x + 50$. What is the slope?
(A) 2
(B) 5
(C) 10
(D) 20
(E) 50
- Q7.** Which equation has an undefined slope?
(A) $y = 2x + 1$
(B) $x = 3$
(C) $y = -4x - 2$
(D) $y = x + 3$
(E) $x = -2y + 1$
- Q8.** What is the slope of the line $x = -2y + 1$?
(A) $-1/2$
(B) $1/2$
(C) -2
(D) 2
(E) 1

Open Ended Questions (FRQ)

- Q9.** A delivery truck travels from City A to City B at an average speed of 60 miles per hour. The distance between the two cities is 210 miles. If the truck leaves City A at 9 AM, what is the equation of the line that models the distance traveled by the truck versus time? Use d for distance and t for time.
- Q10.** The graph shows the relationship between the amount of time t (in hours) a package is in transit and the distance d (in miles) it has traveled. a) Find the slope of the line. b) Interpret the slope as a rate of change. c) If the package travels for 4 more hours, how far will it have traveled?

Chef's Tips